

Diabetes Technology Expert Program

Kaleido

1. Introduction

Welcome to this module on the Kaleido insulin pump! In this module, we will explore the features and operation of the Kaleido insulin pump.

This modular insulin pump can be used as a separate standalone pump, but also in an automated insulin delivery system such as DBLG1 or DBLG2 from Diabeloop with Kaleido. This module specifically focuses on the Kaleido insulin pump, using the dedicated Kaleido handset, providing valuable insights into its usage and functionality.

The Kaleido pump is a semi-patch pump operated using a handheld device, and known for its wide range of color options. Each user is provided with two pumps and can select from ten different color combinations for their pumps and handset, offering choices from vibrant orange to classic black.

Both the handset and pumps are rechargeable and reusable for a period of four years. However, it's important to note that the insulin reservoir, or cartridge, and infusion set should be replaced at least every three days.

One of the standout features of the Kaleido pump is its compact size, making it one of the smallest durable pumps available. It can be attached to the skin using a short tube measuring 5 cm (or 2 inches), or it can be conveniently carried in a pocket with a longer tube of 30 cm (or 12 inches).

In the general module on insulin pumps, you have already learned about the fundamentals of insulin pumps, such as their operation, setting basal and bolus insulin, changing insulin and infusion sets, and managing alarms.

In the upcoming videos, we will showcase Kaleido in action, focusing on specific aspects of the onboarding education for this semi-patch pump. Please be aware that the content presented here may change without prior notification. The User Manual for the medical device remains the primary document for ensuring the safe operation, and it's important to note that these videos are not intended as medical advice. Always consult your healthcare provider before making any adjustments to your therapy.

These videos have been created by Dr. Inge Van Boxelaer, the founder of Diabetotech and an endocrinologist at the St-Lucas Hospital in Ghent in Belgium. The content has been reviewed by Dr. Sarah Siegelaar, an endocrinologist at Amsterdam UMC in the Netherlands, as well as Vanessa Buitendijk from the Netherlands, an individual living with type 1 diabetes who uses the Kaleido pump. Their contributions ensure that the information provided is unbiased and independent.

By the end of these videos, you will have a comprehensive understanding of how the Kaleido insulin pump works, how to set basal and bolus insulin, how to change the insulin reservoir and infusion set, and what actions to take in case of an alarm.

Whether you are a caregiver looking to guide individuals using Kaleido, or a user preparing for a successful start-up, these videos will equip you with valuable knowledge, along with practical tips and tricks to enhance your experience with Kaleido.

We wish you the best of luck on your journey with the Kaleido insulin pump!

2. Pump Operation

Welcome to this video on how the Kaleido insulin pump works! In this segment, we will explore the parts and buttons of the pump and handset, how to charge them, how to unlock the handset, the home screen and menu of the handset, and some do's and don'ts.

2.1 Parts & buttons

The Kaleido insulin pump consists of two main components: the pump and the handset, which controls the insulin delivery of the pump.

The pump is very small, weighing only 19 grams. It consists of a durable and colorful pump base that can be recharged and reused for 4 years, and a disposable insulin cartridge that can be used for 3 days. There are no buttons on the pump, only a space for the insulin reservoir, and a groove for the tubing. The cartridge has to be manually filled with insulin. The insulin cartridge comes with the tubing attached, which can be a short tubing to place the pump on your body, or a long tubing to keep the pump in your pockets. The infusion sets come in separate packaging, and should be connected to the tubing connector.

The handset is also small and weighs only 60 grams. On the bottom there is a micro USB port and a speaker. The micro-USB port is used to charge the

rechargeable lithium battery and for data upload. The handset has a power button on top, and a color screen in the front. Below there are 4 buttons to navigate up and down, and left and right. And in the center there is an “enter” button. On the back of the handset, you can find specifics of the handset like the serial number.

The handset should be kept with you at all times, as it is not only necessary for delivering boluses but also for responding to alarms. Please note that there are no buttons on the pump, so control must be managed using a separate device. The handset communicates with the pump via Bluetooth and needs to be within a range of 2 meters (or 6.5 feet) to establish a connection. Once programmed, the pump does not require continuous communication with the handset for basal insulin delivery.

2.2 Charging the pump base and handset

It is advisable to charge the handset using the provided cable and charger. The optimal practice is to charge the battery each night while you sleep, ensuring a fully charged battery for approximately two days of use. When charging for the first time or when the battery is completely depleted, it may take up to 4 hours to reach a full charge.

When the battery is fully drained, the handset will power off, but the pump will continue delivering basal insulin. You can continue to use the handset while it is charging.

For each of your Kaleido pumps, which you wear for maximum three days at a time, it's a good practice to charge them promptly after removal so they're ready for your next use. This can be done by placing an empty pump into the dedicated charging dock, using the same charging cable as for your handset. An orange light will indicate the charging process. Once your pump is fully charged, the status light on the dock will change from orange to green, signifying that your pump is ready for use. Charging the pump takes about 2 hours.

To remove your pump from the charging dock, simultaneously press the release buttons on both sides of the charging dock – there are two buttons in total. This should release the pump without requiring excessive force.

2.3 Unlocking the handset

You can send your handset to sleep mode at any time by briefly pressing the Power button. This will help you save your handset battery. You can wake it up again at any time just by pressing the Power button or the Enter button.

It is also possible to set a passcode for the lock screen and an emergency message via the Systems Settings. This message functionality allows you to display the name and phone number of your emergency contact, or you can input your own name and phone number so that individuals can reach you if your device is lost.

2.4 Home screen and pump menu

The home screen is divided into five sections, each of which leads to a different menu. At the top, you'll also see handy details like the time and how full your insulin cartridge, pump and handset batteries are.

The 5 sections are Basal, Bolus, Resource Levels, Menu and Status.

- Once everything is set, your Basal section will show the basal rate you are currently receiving. If you need to change or review your basal profile, just navigate to Basal and select it by pressing the Enter button.
- Your Bolus section shows you the details of the last bolus you delivered – that's the date, time and the amount of insulin you got. When you need to deliver your next bolus, navigate to Bolus using the buttons and press Enter.
- If you select the Resource levels, you can see the level of your handset battery, pump battery, amount of insulin in the cartridge, insulin on board, and the maximum time until the next insulin cartridge placement.
- Via the Menu, you can check your History, do a data upload and check and edit Personal Information and System Settings. Although the handset stores up to 3 months of data, the History only shows the data of the past 8 days.
- To pause, stop and restart your pump, navigate to Status, press Enter and go from there.

Each menu ends with a back or home option, to navigate to the previous screen or home screen.

2.5 Do's & don'ts

Here are some important do's and don'ts when using the Kaleido insulin pump:

- We recommend you to save the menu overview provided in the downloads for easy reference. As a healthcare provider, it will help you guide individuals through the pump menu remotely.
- Kaleido pumps are waterproof up to 1.5 meters or 5 feet for up to 1 hour.
- The handset is not waterproof, so avoid exposing it to water.
- Take good care of the handset and pumps, as it should last for approximately 4 years.
- The handset does not automatically adjust the time, date, and time zone. You need to manually update the time when transitioning between winter and summer time or traveling to different time zones. This can be done in the system settings under the yellow menu option.

- It's important to note that it is not possible to change your glucose units in the Kaleido handset. There are different handsets for mg/dl and mmol/l. The type of handset you have is marked on the back.

With this information, you now have a better understanding of the main features of the Kaleido system. One of the key advantages is the small size of the pump and handset, and the fact that you can wear it on your body, without the hassle of long tubing. The pump is waterproof, but it is also possible to disconnect the pump and temporarily remove it from your body. However, keep in mind that you will need to carry both the pump and the handset, to be able to control your pump. Knowing this in advance will help you adjust and adapt.

3. Insulin setup

In this video, we will guide you through the process of setting the insulin delivery on the Kaleido handset.

Please note that the insulin will only start running once you have successfully paired a pump with your handset.

We will cover the following topics:

- Programming basal insulin
- Setting up temporary basal rates
- Giving an insulin bolus
- and some extra do's and don'ts.

Let's get started!

3.1 Programming basal insulin

Before you can start using the Kaleido pump, you'll need to set a basal insulin rate on your handset.

- To do this, go to the home screen, scroll to Basal and press Enter to select. Scroll to and select Create new profile.
- Press the left or right buttons to scroll between each hourly segment. You have to set a basal rate for all 24 hours of the day. Press the up or down buttons to select the basal rate recommended by your healthcare provider for each hour. Below the basal profile graph, you'll see details of the hour you're currently setting, along with the total basal insulin dose. When you've entered a basal rate for each hour and your new basal profile is ready, press the Enter button. Press Enter again to select Next.
- Next up, it's time to enter a name for your new basal profile. Use the up or down buttons to scroll to the letter or number you want to use. Press the

right button to move onto the next letter or enter a space. To scroll through the alphabet quickly, just press and hold the up or down button. When you've named your basal profile, press Enter and select Save.

- Your handset will confirm that your basal profile has been saved. Press Enter to return to the Home screen a little quicker.

Many people opt to use different profiles for different situations, for example work day, weekend, sports and illness. Your handset can store up to seven basal profiles at a time.

If you need to check which basal profile your pump is using, you can go to the basal section and select view current profile. Here you can also change your profile, or edit your profile. It's important to note that it is not possible to change a profile that is currently active. To edit the current profile, you'll need to duplicate it first.

To pause your basal insulin delivery, navigate to the Status section and tap "pause pumping," then confirm with "Yes." Your handset will display the amount of missed insulin since pausing. To resume basal insulin delivery, tap "continue pumping." If you opt to "stop pumping," your handset disconnects from the pump. To restart, access the Status screen, where your handset will ask whether you need to fill your cannula or not. Select the appropriate option, and your pump will restart insulin delivery.

3.2 Temporary basal

If you want to temporarily change the amount of basal insulin you're receiving, you'll need to set a temporary basal rate.

- To do this, go to the Basal section via the homescreen, and select Set temporary basal rate.
- Here you can set a basal insulin rate percentage between 0% to 200%. Once you have your chosen percentage, press the Enter button.
- Set the amount of time you'd like your temporary basal rate to run for. You can choose a minimum of 0.5 hours and a maximum of 24 hours. Once you've chosen the amount of time, press the Enter button again.
- Press the Enter button to select Next.
- Your handset will now show you details of the temporary basal rate that you've set. If the details are correct and you'd like to start your temporary basal rate immediately, press Enter to select Deliver.
- A confirmation will pop up. You can press Enter to return to the Home screen a little quicker.
- To help you keep track, the Basal section of the Home screen will show you your temporary basal rate details for as long as the temporary basal rate is running.

It's important to note that you won't receive an alert when your preset temporary basal rate ends and your normal basal profile resumes.

To stop the temporary basal, you can go to the Basal section, and select Stop temporary basal rate.

3.3 Delivering a bolus

You have several options to administer a bolus using the handset. You can utilize the Kaleidoculator, which is an integrated bolus calculator. Or you can opt for a manual Quick bolus, which is a standard bolus, or a manual Extended bolus, which is a dual-wave bolus.

Before you can use the Kaleidoculator, you have to configure it via the Personal Info in the Menu section. There you can set the Target BG, Insulin to carb ratio, Insulin sensitivity, and the Duration of insulin action. For more information about these parameters, please check the general module on insulin pumps.

To deliver a bolus with the bolus calculator, navigate to the Bolus section, and choose Kaleidoculator.

- You will first be asked to enter the grams of carbohydrate you are going to eat. You can enter for example 25 grams and tap Next. If you're not eating, but want to use Kaleidoculator to help correct your BG, you can leave the number boxes empty and press the Enter button to select Next.
- You'll need to enter your current BG reading, then press next. It is also possible to give a bolus without entering a BG-reading. Once you select Next, the BG reading and amount of carbs you've entered will be saved in History.
- The Kaleidoculator will show you your current Insulin on Board, that's how many units of insulin are still active in your body. You can accept or reject the Insulin on Board to be included in your calculation.
- The Kaleidoculator will now show you details of its recommended bolus. You have 2 options to choose from:
 - to deliver the recommendation as a Quick Bolus
 - or to deliver the recommendation as an Extended Bolus.
 - Via the next step you can also adjust the amount of insulin to be delivered.
 - If you decide you don't want to deliver a bolus after all, just scroll left to Cancel. If you still want to give the bolus, tap Next, and then Deliver. The bolus amount that's being delivered is visible on your home screen.

With the Kaleido handset, you can also give a manual bolus without the bolus calculator. To do that, go to the Bolus section and choose Quick bolus. Here you can choose the amount of insulin you want to give and tap Next and then Deliver.

For a manual Extended bolus, you can also choose that option. First you enter the amount of insulin you wish to receive, and the amount you wish to deliver immediately, and finally the duration over which you wish to deliver the remaining bolus.

3.4 Do's & don'ts

Here are some do's and don'ts to keep in mind:

- Ensure your maximum bolus limit is set slightly above your expected bolus doses. You can set this between 0 and 30 units.
- If the Kaleidoculator recommends a bolus that is bigger than the maximum bolus, you'll get a popup that warns you about this. If you'd like to continue with this high bolus, don't forget to deliver another bolus of the remaining units after your first bolus has finished.
- Sometimes, Kaleidoculator may indicate a negative amount of insulin as its recommended bolus for you. If you're not eating or not eating very much carbs, and if your blood glucose is already below your target blood glucose, you don't need more insulin, you actually need less. If you select either Quick Bolus or Extended Bolus when your recommended bolus is a negative number, your handset will pop up with a quick reminder to tell you that giving a bolus isn't the safest thing to do at this time.

In conclusion, setting the insulin delivery on the Kaleido handset is a straightforward and intuitive process. As a healthcare provider, it is crucial to familiarize yourself with the different pumps and their settings. With Kaleido, you can easily access all the necessary settings under the Basal section and you can find the settings for the bolus calculator under the Personal Info in the Menu section.

4. Replacing the insulin reservoir and infusion set

Welcome to this lesson on how to replace the insulin reservoir, or cartridge, and infusion set of the Kaleido insulin pump.

In this video, we will cover the available infusion sets, the 4-step process for changing the insulin cartridge and infusion set, and last but not least, some important do's and don'ts.

It's important to note that if you intend to use the Kaleido pump as an Automated Insulin Delivery system with Diabeloop, you should not pair your pump with the Kaleido handset, but instead with the DBLG handset.

4.1 Different infusion sets

The Kaleido pump has one type of infusion set with a teflon cannula in a 90 degree angle. You can choose between a cannula of 6 or 9 mm.

Additionally, Kaleido provides two lengths of tubing, which are included with the insulin cartridge. Most people use the short tubing (5 cm or 2 inches) to keep the pump close to the infusion site on the skin, but you can also opt for the longer tubing (30 cm or 15 inches) to wear the pump in your pocket.

Below this video you can find an overview and comparison of the available infusion sets of different insulin pumps, their tubing and cannula lengths, and filling volumes.

4.2 Insulin cartridge and infusion set replacement

To replace the insulin cartridge and infusion set, gather the following supplies: your Kaleido handset, a charged pump, a body and pump patch, insulin at room temperature, a new insulin cartridge, a syringe and needle, an infusion set, an infusion set inserter and alcohol wipes.

It's important to note that if you intend to use the Kaleido pump as an Automated Insulin Delivery system with Diabeloop, you should not pair your pump with the Kaleido handset, but instead with the DBLG handset.

#1 Step: Filling the insulin cartridge and the tubing

Start off by washing your hands and preparing a clean flat surface to work on. Be sure to check the expiry date and ensure the packaging and insulin cartridge are not damaged. If there is any damage present or if it's expired, discard and use another one. Start by opening the packaging from your insulin cartridge by slowly pulling from the bottom. Place your insulin cartridge so that the numbers 1 and 2 are facing towards you.

Remove the syringe from its packaging. Without moving the plunger, push the tip of the syringe firmly into Hole 1. Carefully pull the plunger all the way up to the top of the syringe. You should feel some resistance. This will remove any excess air from your insulin cartridge. Now, remove the syringe from Hole 1 and push the plunger back towards the tip of the syringe.

Next, remove your needle from its packaging. Push the needle onto the tip of the syringe until they are connected. Take your room temperature insulin vial, check the expiry date and clean the rubber stopper with an alcohol wipe before use. Remove the needle cap from the needle. Draw the plunger back to fill the syringe with 2.5ml of air. Place the insulin vial in front of you onto the flat surface. Carefully push the needle through the rubber stopper of the insulin vial. Push the plunger all the way in to put all the air from the syringe into the vial. Keep holding the plunger down. Turn the vial with the syringe upside down and slowly pull the plunger down till you've drawn 2.5 ml of insulin. Give the syringe a few taps on the bottom to push any air bubbles up towards the needle. Push the plunger upwards to push the air bubbles back into the insulin vial. Ensure that there is 2.5ml of insulin in your syringe. Check the syringe for air bubbles, some small champagne sized air bubbles are acceptable. Remove the syringe from the insulin vial, be careful not to touch the needle. Turn the syringe so the needle is pointing downwards and give a few firm taps on the syringe. This will ensure that any air bubbles that remained in the syringe go upwards near the plunger.

Holding your insulin cartridge, gently place the needle into Hole 2. Slowly push down the plunger, filling your cartridge with insulin. You must fill your insulin cartridge completely. Stop pushing the plunger when you see a drop of insulin coming out of Hole 1 (stop before any air bubbles get transferred into the insulin cartridge). It is normal that some excess insulin remains in the syringe after filling the insulin cartridge. Remove the syringe and dispose of the needle and syringe into a sharps bin.

Now you will need to prime and then remove your insulin cartridge from the filling cradle. First, hold the top of your insulin cartridge on both of the round edges and squeeze it further into the filling cradle. This will unhook the plastic tab. Then squeeze downwards fully into the filling cradle twice and then release the insulin cartridge. It is important that this action is done twice. You can now remove your insulin cartridge from the filling cradle. Turn your insulin cartridge over. Squeeze the tabs on either side of the insulin cartridge connector and then lift up to unclip it.

Finally, have one last check to make sure there are no large air bubbles in the reservoir. If large air bubbles are present, discard and use another insulin cartridge. If they are champagne sized air bubbles, that's generally okay and you can use this insulin cartridge. If you see any air bubbles in your insulin cartridge, check your User Manual for what is acceptable or what is not. Your insulin cartridge is now fully filled, primed and ready to use! Now it is time to prepare and wear your pump.

#2 Step: Infusion set placement

You'll need to consider where your pump will be placed in relation to your infusion set. Start by washing your hands and choose where you'd like to position your infusion set. Prepare the area by cleaning it with an alcohol wipe. Allow your skin to dry naturally. Check the expiry date and ensure the packaging and infusion set are not damaged or expired before use. Peel open the paper lid. Remove the protective circle of plastic that's sitting on top of your infusion set. Press the button on the top of your inserter to make sure it's ready to use. Take your inserter and line it up with the top of your infusion set. You can do this by making sure the gap in the base of the inserter sits above the tabs on the infusion set's plastic cap. Push the inserter firmly onto the top of your infusion set. Push down until you hear the Infusion set click into your inserter. Lift your inserter out of the packaging and check if the needle is straight. Be careful not to touch the needle! It's important that it stays sterile. Peel the paper backing off the sticky part. Try not to touch the sticky area. If possible, use one hand to hold your skin taut of the area where you want to place your infusion set. Use the gap in the base of your inserter as a guide for where your insulin cartridge connector will connect to your infusion set. Use your other hand to position your inserter on your skin. When your inserter is in the right place, firmly press the button at the top of your inserter. Keep the inserter pressed firmly onto your skin to aid the insertion. This will push the infusion set into your body. Now pull your inserter away from your body in a straight line without twisting it. Put the inserter aside for now. Be careful not to touch the needle. Gently rub the adhesive plaster around the edge of your infusion set to make sure it's sticking to your skin properly. If the Infusion set is peeling from the skin at this stage, it's unlikely that it will last for three full days of wear. If this is the case, you should remove it and try again with a new one. Now, you need to dispose of your needle. Don't forget to keep your inserter safe, ready for next time!

#3 Step: Preparing your pump

Turn your handset on. Have your fully filled insulin cartridge ready to go. Hold your pump with the pump's white parts facing towards you. Hold your insulin cartridge above the space and line it up, tipping it to feed the tubing into the cartridge tubing groove. Gently press the cartridge tab to feel it click into place. Never push your insulin cartridge straight in or put excessive force on it. This can damage the pump and cartridge. When inserted correctly, you will hear 2 beeps to signal your pump is now ready to be connected to the handset. If everything's good, the back of the insulin cartridge will sit flush with the back of the pump.

Only if this is the first time you use this pump, a message will pop up on your handset, letting you know your pump is ready to pair. Make sure your pump color and serial number matches the pump you are trying to pair. When you are ready to

pair, press Enter to select Pair. The pairing process may take up to a minute. Once pairing is successful, the Handset will show a confirmation message. The pump will beep three times to let you know pairing was successful. If you have already used and paired your pump previously, your Handset will ask you to connect to your pump. Press Enter to select connect and now your pump is connected.

Now your handset will ask if you have used a new, fully filled insulin cartridge. If you just filled your insulin cartridge, press Enter to select Yes.

It is time to apply your patches! You will need your pump and body patch. Take your pump patch and peel off the paper backing. This will reveal the patch's adhesive. Stick the pump patch to the bottom of your pump, so that it covers the insulin cartridge and the fuzzy hexagon shape is facing outwards. Now take your body patch and without removing its paper backing, stick it to your pump patch by pressing together the hexagon shapes. Make sure the body patch is aligned with your pump.

Support the weight of your pump, and clip the connector at the end of your insulin cartridge tubing into your infusion set. You'll know it's secure when you hear the two clicks as the right and left hand sides connect. Check whether the connector is smoothly aligned with your infusion set. You should not feel bumps or gaps. Holding onto your pump, peel off the paper backing from the top of your body patch. Now it is time to place your pump where you'd like to wear it. Be careful not to stretch or kink your insulin cartridge tubing. When you are using the 30 cm or 15 inch insulin cartridge tubing, place your pump in a secure place. If you are using a new infusion set, you need to prime it first. Select the cannula length you are using and press Enter. Press Enter again to select Yes to start pumping. Your cannula is now being primed. Once priming has completed, your handset will confirm that pumping has started. Your pump is now pumping!

#4 Step: Removing your pump

If your insulin reservoir is empty or when you have used the pump for 3 days, you'll have to stop and remove your pump. You can do that by going to the status section and tap stop pumping. You'll receive a confirmation.

Then you can remove your pump. Start by washing your hands. Gently squeeze the tabs on both sides of your insulin cartridge connector. Lift it up to unclip it from your infusion set. Lift the pump away from your body patch. Next, gently remove the body patch from your skin, as well as the patch from the back of your pump. Take your insulin cartridge out of your pump by pulling on the tab. And finally, remove

your infusion set. Dispose of your infusion set, insulin cartridge, body and pump patches according to local regulations.

Place your used pump on the charging dock. It will take about 2-4 hours to fully charge your pump.

4.6 Do's & don'ts

Here are some important points to keep in mind:

- You should replace your insulin cartridge when it's empty or after 3 days of use. This is also a good time to switch between your two pumps for recharging. The infusion set should also be changed at least every 3 days. Ideally, you can do everything together, but if needed, you can replace the parts separately. If your insulin cartridge empties before 3 days, or your pump battery is empty, you can do a separate replacement more quickly.
- The Kaleido pump is compatible with Novorapid and Humalog with a concentration of 100 U/ml.
- The cartridge always has to be filled with 200 units of insulin.
- Do not reuse insulin reservoirs. Doing so may contaminate your insulin, lead to infection, and possible under or no delivery of insulin.
- Do not reuse body and pump patches or use them longer than three days. Doing so may lose their adhesive properties, which could result in your pump becoming detached during use.
- Make sure that the back of your pump is clean and dry before you attach a pump patch. This will ensure a better bond between the patch and pump, and prevent your pump from coming loose.
- Do not pair your pumps to your Kaleido Handset if you intend to use the system as part of the Diabeloop System.
- Once you've placed your pump where you want it, try not to play with it too much or bring it into contact with water during the first hour of wear. This will make sure it's stuck securely.
- Remember to check your blood glucose levels 2 hours after inserting a new infusion set. This is the only way to confirm proper insertion and insulin delivery. It's best to avoid changing your infusion set right before bedtime.
- When temporarily removing your insulin pump, it's best to close the protective cover on your infusion set. If you leave the protective cover open, dirt may enter into the opening which may cause infections or occlusions.

That's it! Now you're familiar with the process of replacing your Kaleido pump and infusion set. We've personally found wearing the pump quite comfortable, hardly noticing its presence. However, if the cannula hits a muscle, it could cause discomfort, in which case it's best to insert a new infusion set. If you're unsure about

the replacement steps, clear explanatory videos are available on Kaleido's YouTube channel. Especially in the beginning it is helpful to use these videos when you replace your insulin cartridge and infusion set.

5. Pump alarms

Welcome to this lesson about the alarms on the Kaleido insulin pump.

The handset and or pump can alert you about various issues through beeping or vibrating. Kaleido classifies these alerts as either low or high priority alarms.

If multiple alarms happen simultaneously, the intelligent alarm system will first notify you of high priority alarms, followed by low priority ones. Each alarm must be acknowledged individually before the next one appears. The alarm sounds and vibrations will repeat until you acknowledge the message on your handset. To do so, simply press the Enter button after reading and understanding the alarm message.

Let's delve into these alarms and share some extra tips.

5.1 High priority alarms

High priority alarms demand immediate attention. They will notify you with 10 short beeps on your pump or handset and will stop your pump if it's related to the pump, or turn off your handset if it's related to it.

You'll get a high priority alarm related to the pump if:

- Your pump battery is empty and infusion stopped
- Your insulin cartridge is empty
- The insulin cartridge you're using has expired
- Your pump has an insulin blockage
- Your insulin cartridge has come out of your pump while it was pumping
- There's a problem with your Kaleido pump

You'll get a high priority alarm related to the handset if:

- There's a problem with your Kaleido handset

5.2 Low priority alarms

Low priority alarms are important but not urgent. They notify you with 2 or 3 short beeps and won't stop your pump or handset.

You'll get a low priority alarm related to the pump if:

- Your pump battery is getting low
- You've been using the same insulin cartridge for three days
- Your insulin cartridge volume is getting low

You'll get a low priority alarm related to the handset if:

- Your handset battery is getting low
- There is a communication error between your handset and your pump
- Your Insulin on Board has been reset
- Your handset battery is empty
- Your handset and pump software versions do not match

5.3 Do's & don'ts

Here are some important points to remember:

- You should always keep your handset within reach in case there's an alarm and keep the handset battery sufficiently charged.
- Old alarms can be reviewed in the History Menu. Active alarms will be visible if you wake your handset from sleep mode.
- If you are unsure how to respond to an alarm, consult the user manual or contact customer service.
- For some pump and handset alarms, the handset will show a 4 digit number in the bottom right corner. The alarm number can also be found in your History on the handset and in the manual. In case you are not able to resolve the issue, contact customer service and they will ask you for the 4 digit number.
- If you temporarily won't be using your handset, you can simply switch it off. To do this, hold the power button until you can confirm to turn off the handset.

Congratulations! You have completed the final lesson on the Kaleido pump. Initial setup is usually supervised by a diabetes nurse, and if you have any further questions, don't hesitate to consult them. Well done on reaching this point!

6. Case report

Welcome to the final part of this Kaleido insulin pump module.

Now that you have explored the key features, setup process, and daily use of the Kaleido pump, we invite you to learn from the experiences of people who use the system in real life.

First, we recommend watching an independent review moderated by Justin from Diabetech. In this longer discussion, users share practical advantages and limitations of the Kaleido pump, including its colourful design options, tubing choices, wearability, and day-to-day experience. The first part of the video focuses on the pump itself, while the second part discusses the DBLG1 automated insulin delivery system, which has since evolved into the newer DBLG2 system.

Video link: <https://www.youtube.com/watch?v=KlchDGuH3OI>

We also recommend watching a short video demonstrating a complete infusion site change in less than two minutes. One particularly useful tip shown in the video is how the user first attaches both adhesive patches to the pump before applying the entire assembly to the body. This simple technique can make the insertion process quicker and easier, as there is no need to align the patches afterwards.

Video link: <https://www.youtube.com/shorts/k9TvRbrc2nQ>

Together, these videos provide valuable practical insights that complement the information presented in this module.

Congratulations on completing the Kaleido pump course. Your commitment to learning about diabetes technology will help you make informed decisions and support others using these systems. As always, continued learning, experience, and collaboration with healthcare professionals remain essential for achieving the best possible outcomes.

We wish you every success with the Kaleido system and encourage you to continue exploring, learning, and sharing experiences with the diabetes community.